Coding 2

Python for Data Analysis

ECBS5306 Central European University

Ulrich Wohak 2 November 2026

Welcome and course information

Programming and analytical judgment

Python skills you bring

- Variables, collections, functions
- Conditions, loops and file paths
- pandas tables and plotting
- Git, GitHub and uv

Questions we will answer

- How do variables relate?
- How well does it predict new data?
- What makes a result credible?

Our teaching sessions

Six Mondays, 15:40–17:20

2 November to 7 December 2026

- We build and discuss the notebook together.
- You each develop your own prompt for an LLM to reconstruct our analysis.
- We compare the responses with our executed notebook.

Bring your laptop. Type the code, predict results, and ask questions.

Assessment

40%

Debugging homeworks

Six homeworks, one per session
Equal weight within the 40%
Completed outside class

60%

Final exam

Written and closed-book
No computer
Covers the entire course

Passing threshold: 50/100 overall

The registrar will announce the exam slot.

Debugging homework

You review a small analysis written by a colleague.

- Each assignment has its own GitHub repository and uv environment.
- Read the README, reproduce the problem, and investigate the analysis.
- Correct the code and check whether the conclusions follow from the results.
- A notebook can run successfully and still give a misleading answer.

Homework submission and marking

- Create your own copy of the released repository and run uv sync.
- Repair the analysis and complete its interpretation prompt.
- Restart the kernel and run the final notebook from top to bottom.
- Commit and push the completed work using the assignment’s instructions.

What the grade reflects

Reproducibility, correct repairs, and an interpretation supported by the evidence. You must be able to explain your changes.

AI use and responsibility

AI assistance is allowed for debugging homeworks.

- Use it to understand an error or propose a repair.
- Check each suggestion against the data and official documentation.
- Run the complete workflow and verify the interpretation yourself.
- Be ready to explain every submitted line and conclusion.

AI tools are prohibited during the closed-book final exam.

The classroom reconstruction

- First, we complete and run our own analysis.
- You each use that workflow to develop your own prompt for an LLM.
- We compare the sample, steps, vocabulary, results, and interpretation.
- We investigate omissions and decide whether differences matter.

Our executed notebook is the reference for the comparison.

The six-session roadmap

Date	Topic	Homeworks
2 Nov	Introduction and simple linear regression	1
9 Nov	Features, interactions and diagnostics	1
16 Nov	Logistic regression	1
23 Nov	Validation, resampling and regularization	1
30 Nov	PCA and clustering	1
7 Dec	Regression trees and course synthesis	1

One homework per session: six homeworks in total.

Data and reading

Data Analysis for Business, Economics, and Policy

Gábor Békés and Gábor Kézdi

- European hotels: regression and diagnostics
- SHARE health: binary outcomes
- Airbnb listings: validation and regularization
- World Management Survey: PCA and clustering
- Used cars: regression trees

The course repository contains the required notebooks and data setup.

Course setup

Clone the course repository and open its folder.

github.com/ulrichwohak/Coding-2-Python-for-Data-Analysis

```
cd Coding-2-Python-for-Data-Analysis
uv sync
uv run python scripts/fetch_data.py
uv run jupyter lab
```

Run these commands from the project folder. Use its Python environment.

Notebook execution and documentation

- Use the Python kernel from the project’s uv environment.
- Run cells in order and inspect the data after important steps.
- Read error messages and consult official documentation.
- Restart the kernel and run all cells before calling the analysis complete.

Check what each method accepts, returns, and assumes.

Today: Vienna hotel prices

How do hotel prices vary with distance?

- Inspect the sample and the units: USD and miles.
- Plot prices against distance and fit a simple regression.
- Interpret the fitted intercept and slope.
- Compare observed prices with the fitted line.

Our simple regression describes an association in the observed data.

Today’s starting notebook

Simple linear regression

`lectures/lecture12-simple-linear-regression/`

`simple_linear_regression.ipynb`

Later: multiple regression and workflow

`lectures/lecture13-multiple-regression-workflow/`

`multiple_regression_workflow.ipynb`

Today’s homework covers simple linear regression.

Reference: CEU AI in the terminal

For the class reconstruction and permitted homework assistance

```
uv run ceu.py
```

- Run it in a terminal inside the course folder.
- Use the access details supplied separately by the instructor.
- Inspect generated files, run them yourself, and check the results.
- Keep access credentials out of notebooks, prompts, and Git.

Reference: materials and help

Course materials

github.com/ulrichwohak/Coding-2-Python-for-Data-Analysis

Official documentation

pandas.pydata.org/docs/
www.statsmodels.org/stable/
scikit-learn.org/stable/

Course questions

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